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BACHELOR THESIS

By

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Title:

Development of an Internal AI Chatbot for Company Knowledge Retrieval

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Internal Supervisor: Dr. Trần Hoàng Tùng

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To whom it may concern,

I, **Trần Đình Chí** (supervisor’s name), certify that the thesis/ internship report of Mr/Ms. **Nguyen Duy Ngoc** (student’s name) is qualified to be presented in the Internship Jury 2025–2026.

Hanoi, ___/___/______

Supervisor’s signature

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
BM25	Best Match 25
CPU	Central Processing Unit
GGUF	GPT-Generated Unified Format
GPU	Graphics Processing Unit
HR	Human Resources
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NLP	Natural Language Processing
QLoRA	Quantized Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
RRF	Reciprocal Rank Fusion
UI	User Interface
USTH	University of Science and Technology of Hanoi
VRAM	Video Random Access Memory

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Abstract

This thesis presents a fully local Retrieval-Augmented Generation (RAG) chatbot for Vietnamese Human Resources policy questions. Twenty policy documents are indexed with multilingual sentence embeddings and ChromaDB. A hybrid retriever combines dense similarity and BM25 rankings through Reciprocal Rank Fusion, then supplies the highest-ranked passages to a Phi-3-Mini GGUF responder. On a small, manually labelled 10-query retrieval test, hybrid retrieval reached 100% top-1 accuracy compared with 60% for the dense baseline.

The study also investigates domain adaptation. Qwen2.5-72B-Instruct generated 4,900 unique question-answer pairs across the 20 policy topics. A keyword-based validation classified 82.4% as clearly HR-related, with no exact duplicate questions or predefined off-domain red flags. QLoRA training on an H200 GPU completed three epochs in 5.3 minutes. On 20 direct, retrieval-free questions, the fine-tuned model increased an automated format/relevance score from 0.62 to 0.70 and reduced mean latency from 3.9 to 0.9 seconds. However, manual inspection still found incorrect policy values and incoherent answers; therefore, the automated score is not evidence of factual accuracy. The final system retains RAG grounding, while fine-tuning is treated as complementary. The results show that hybrid retrieval improves short-query matching and that dataset and evaluation quality remain decisive for reliable local domain assistants.

Keywords: Retrieval-Augmented Generation, Vietnamese NLP, BM25, Reciprocal Rank Fusion, QLoRA, local language model.

1. Introduction

1.1 Context and Problem

Company policies are frequently distributed as long PDF handbooks. Employees must locate rules on leave, salary, insurance, remote work, or disciplinary procedures, while HR staff repeatedly answer similar questions. Keyword search helps only when users know the exact wording used by the document. The problem is harder in Vietnamese because short colloquial questions may omit formal policy terms.

Cloud language models offer fluent interaction but can expose confidential documents, require network access, and introduce recurring cost. A local system avoids these constraints, yet a small model cannot be expected to memorize each company's policy or reliably infer facts from its parameters. RAG addresses this limitation by retrieving relevant passages at query time and conditioning generation on those passages [1].

This work asks whether a compact, offline pipeline can retrieve Vietnamese HR policy passages accurately enough to support a local responder. It also asks whether QLoRA domain adaptation improves a small model when trained on a quality-gated synthetic dataset. The second question is deliberately separated from deployment: an ungrounded model may produce plausible but incorrect policy values even when its language and speed improve.

1.2 Related Work

RAG combines non-parametric retrieval with neural generation, allowing evidence to be updated without retraining [1]. Dense retrievers encode queries and passages into a vector space. Sentence-BERT made this practical for semantic similarity [5], while multilingual encoders extend the approach across languages.

BM25 remains a strong lexical ranking method [2]. Dense retrieval captures meaning but may miss exact terms in short queries; BM25 has the opposite strengths. Reciprocal Rank Fusion (RRF) combines ranked lists without requiring calibrated scores [3]. This makes it useful when vector similarity and BM25 scores have different scales.

Phi-3-Mini is a compact instruction-tuned model intended for constrained local deployment [4]. LoRA adds trainable low-rank matrices while freezing base weights [7]; QLoRA applies this method to a quantized base model to reduce training memory [6]. These techniques make adaptation affordable, but they do not guarantee that synthetic labels are correct or that automatic relevance checks measure factual accuracy.

Vietnamese introduces additional retrieval difficulty because whitespace does not always correspond to word boundaries. PhoBERT illustrates the value of language-aware pretraining for Vietnamese NLP [10], while ViQuAD provides a reference point for Vietnamese machine-reading evaluation [11]. The present implementation uses a multilingual encoder and simple whitespace BM25 tokenization, which is efficient but remains a limitation.

1.3 Contributions

The main contributions are:

1. a private, offline Vietnamese HR RAG pipeline using local embeddings, ChromaDB, hybrid retrieval, and Phi-3-Mini GGUF inference;
2. a comparison of dense and hybrid retrieval on short Vietnamese queries;
3. a quality-gated pipeline producing 4,900 unique Vietnamese HR Q&A pairs;
4. a measured H200 QLoRA experiment with an explicit distinction between automated relevance indicators and factual policy correctness.

1.4 Research Questions and Scope

The evaluation is organized around three research questions. **RQ1** asks whether combining lexical and dense rankings improves top-document retrieval for short Vietnamese HR questions. **RQ2** asks whether a controlled synthetic dataset can shift Phi-3-Mini toward Vietnamese HR language and reduce direct inference latency. **RQ3** asks whether the available automated indicators are sufficient to establish factual policy correctness.

RQ1 is addressed by a paired comparison on the same ten labelled queries. RQ2 uses identical 20-question prompts for the base and adapted models. RQ3 requires manual inspection of those outputs and analysis of the scoring code. The scope does not include a legal assessment, deployment at a real company, or a formal human-subject study. This boundary is important because a technically functional chatbot and a validated HR decision-support system are not equivalent.

2. Objectives

The project has four objectives:

1. design and implement a fully local Vietnamese HR question-answering system whose source documents remain on the user's machine;
2. compare dense retrieval with BM25–vector fusion for short HR queries;
3. build and validate a synthetic domain dataset, then measure QLoRA adaptation of Phi-3-Mini;
4. evaluate the system transparently, including incorrect outputs and limitations of small test sets and heuristic metrics.

The deployed architecture is successful when it retrieves a relevant policy source, constructs a grounded prompt, and returns a Vietnamese answer with a source reference. The fine-tuning experiment is evaluated separately because its direct, retrieval-free test measures model adaptation rather than end-to-end RAG correctness.

3. Materials and Methods

3.1 System Architecture

Figure 3.1 shows the offline processing and query paths. At indexing time, Vietnamese policy documents are split into overlapping chunks, embedded, and stored in ChromaDB. At query time, vector and BM25 retrieval run over the same corpus. RRF selects the top passages, which are inserted into a Phi-3 chat prompt. The model runs through `llama-cpp-python`; no company document is sent to an external inference API.

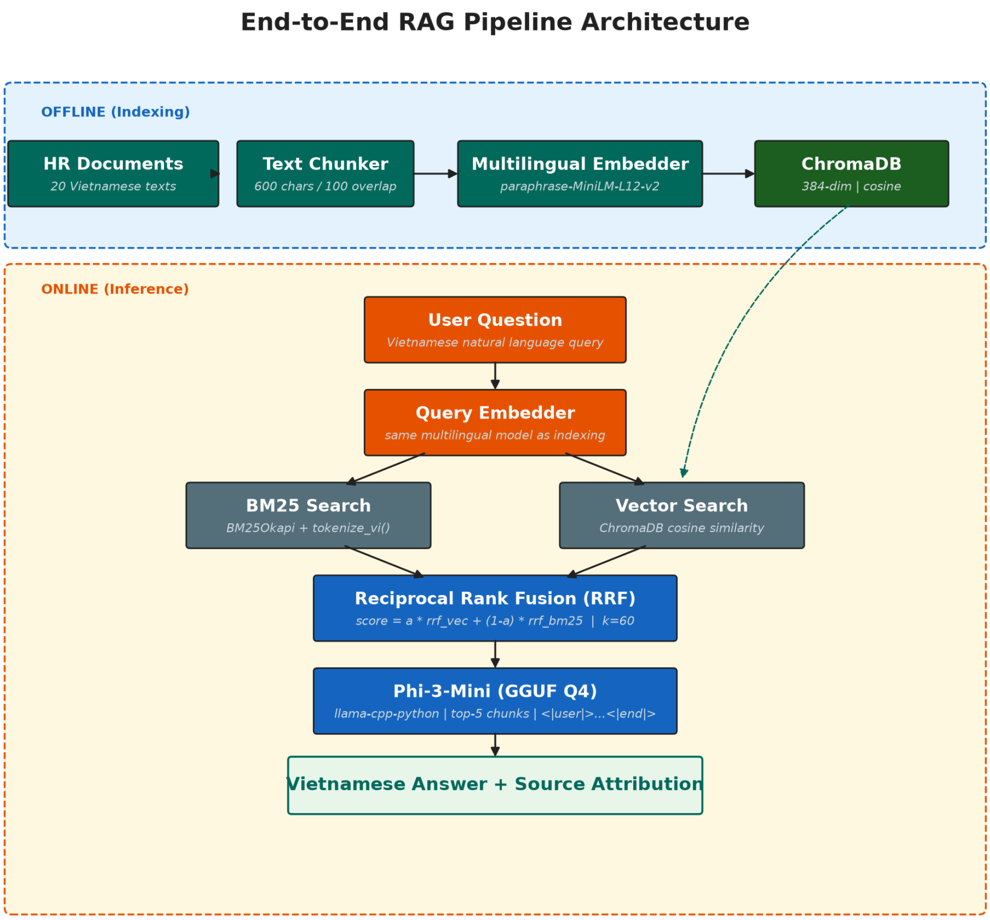


Figure 3.1: *Offline Vietnamese HR RAG architecture.*

Table 3.1: *Principal implementation components.*

Stage	Technology	Role
Documents	20 UTF-8 policy files	Vietnamese HR knowledge source
Chunking	Recursive splitter, 600 characters, 100 overlap	Preserves local context
Embedding	multilingual MiniLM, 384 dimensions	Dense semantic representation
Vector store	ChromaDB	Persistent cosine search
Lexical search	BM25Okapi	Exact-term ranking
Fusion	RRF, $k = 60$	Combines vector and lexical ranks
Generation	Phi-3-Mini Q4_K_M	Local Vietnamese response generation

3.2 Knowledge Base and Indexing

The knowledge base covers 20 topics, including leave, salary, contracts, social insurance, maternity, overtime, remote work, IT security, workplace safety, and anti-harassment. The Vietnamese Labour Code provides general legal context for these topics [9], but the demonstration policies are not legal interpretations. Eight initial documents were manually authored and twelve were generated to broaden topic coverage. The corpus represents a demonstration company rather than the policies of a real employer; answers must therefore not be interpreted as legal advice.

After cleaning, the documents produced 286 stored chunks. Each chunk retains its source name for attribution. The multilingual MiniLM encoder maps text into 384-dimensional vectors stored in a persistent ChromaDB collection [8]. Figure 3.2 summarizes corpus coverage.

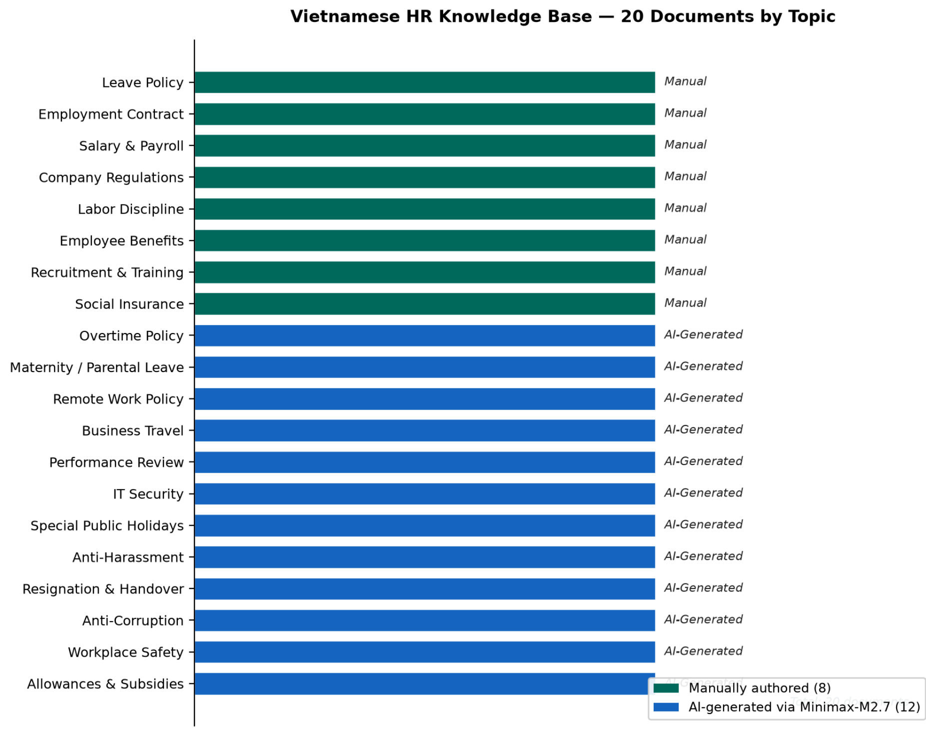


Figure 3.2: *Topic coverage of the 20-document knowledge base.*

3.3 Hybrid Retrieval

Dense search retrieves semantically related text, while BM25 emphasizes query terms. Their ranked lists are fused using

$$S(d) = \alpha \frac{1}{k + r_v(d) + 1} + (1 - \alpha) \frac{1}{k + r_b(d) + 1}, \tag{3.1}$$

where r_v and r_b are the vector and BM25 ranks, $k = 60$, and $\alpha = 0.5$. The top five fused chunks form the generation context. Because RRF uses ranks rather than raw scores, no cross-system score calibration is needed.

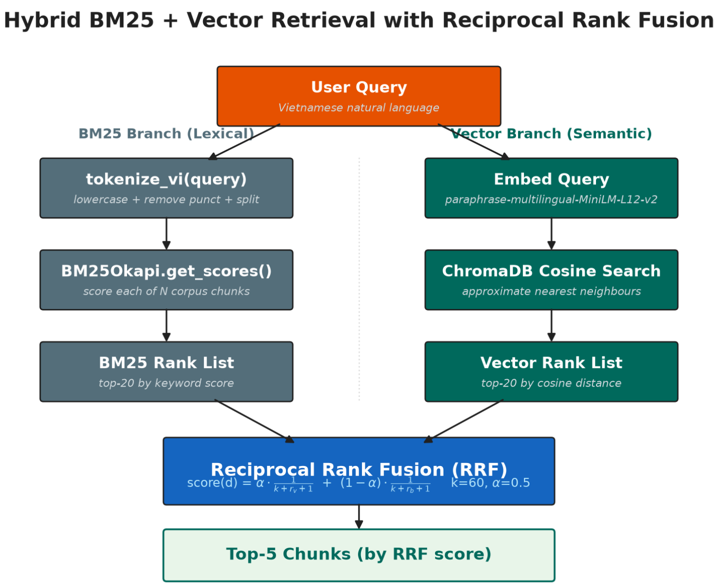


Figure 3.3: *BM25 and vector rankings combined through RRF.*

3.4 Answer Generation

The responder is Phi-3-Mini-4k-Instruct in GGUF Q4_K_M format [4]. The prompt contains a Vietnamese instruction, the retrieved chunks with source labels, and the user question inside Phi-3 chat tokens. Generation uses low temperature and stops at the model's chat boundary. Source grounding is achieved by the prompt and retrieval context, not by an assumption that the model's parameters contain correct company rules.

3.5 Implementation and Privacy Controls

The implementation separates ingestion, retrieval, and generation into Python modules. PDF extraction and chunking create serializable records; embedding code owns the ChromaDB collection; the hybrid retriever reads the same chunk set for BM25 and vector ranking; and the responder wraps `llama-cpp-python`. This separation permits retrieval tests without loading the 2.32 GB generator and makes the source of a failure easier to identify.

At startup, the application checks that the GGUF and vector collection exist. A user question is normalized, embedded, and sent to both retrieval branches. The fused top-five passages are formatted with their source names before local generation. The interface returns the generated text together with source metadata so the user can inspect the underlying policy rather than treating the model output as an authoritative decision.

Privacy follows from local data flow: policy text, embeddings, prompts, and answers remain on the machine during normal operation. The generation pipeline used external compute only to create the synthetic research dataset and train the adapter; it did not form part of deployed inference. A real organization would additionally require access control, encrypted storage, audit logging, retention policy, and review of uploaded documents. These controls are outside the prototype but are necessary before production use.

The principal failure control is abstention through prompt instruction when the retrieved context is insufficient. This is weaker than formal entailment. Source display therefore remains important, especially for numerical benefits or legal questions. No model response should replace an approved policy document or HR professional judgement.

3.6 Synthetic Data Generation and Validation

An initial fine-tuning attempt used a web-crawled dataset that mixed unrelated material with HR content. Its outputs included references to Medicare and other off-domain subjects. Rather than treating that run as representative of QLoRA, the experiment was repeated with data generated only from the 20 controlled HR documents.

Qwen2.5-72B-Instruct-AWQ ran with vLLM on a Vast.ai H200. Prompts requested natural Vietnamese questions and source-grounded answers for each document. Generation resumed by document and batch, normalized questions were deduplicated, and the accepted output stopped at the available final total of 4,900 pairs.

Validation checked JSON structure, length, exact normalized question duplicates, predefined off-domain red flags, and a Vietnamese HR keyword set. These checks are reproducible but shallow: a pair labelled “unclear” may still be relevant, and a pair containing an HR keyword may still contain a wrong policy value.

Table 3.2: *Validation summary for the H200-generated dataset.*

Measure	Result
Accepted Q&A pairs	4,900
Source documents	20
Keyword-classified HR-relevant	4,037 (82.4%)
Unclear under keyword rules	863
Predefined red flags	0
Exact duplicate questions	0
Short questions ($<$ 10 characters)	4
Short answers ($<$ 10 characters)	65

All 20 source topics remained represented after deduplication. The distribution is not uniform because document length and parseable model output differed by batch. Table 3.3 records the final counts and makes this imbalance explicit rather than treating 4,900 as a uniformly stratified sample.

Table 3.3: *Accepted Q&A pairs by source topic.*

Topic	Pairs	Topic	Pairs
Social insurance	122	IT security	230
Remote work	327	Salary policy	184
Leave policy	264	Business travel	160
Performance review	254	Employment contracts	262
Labour discipline	256	Overtime	240
Special leave	287	Maternity	242
Resignation/handover	287	Company regulations	338
Anti-corruption	179	Allowances	181
Employee benefits	282	Anti-harassment	300
Workplace safety	219	Recruitment/training	286
		Total	**4,900**

3.7 QLoRA Training and Direct Evaluation

The second run adapted `unsloth/phi-3-mini-4k-instruct-bnb-4bit` using QLoRA. Table 3.4 records the configuration from the training log. The LoRA adapter was saved, merged, and recovered as a 2.32 GB HR-labelled Q4_K_M GGUF. The artifact's metadata identifies the fine-tune as `hr`.

Table 3.4: *H200 QLoRA training configuration and outcome.*

Parameter	**Value**
Examples / epochs	4,900 / 3
Sequence length	2,048 tokens
Device batch / accumulation	8 / 2 (effective 16)
LoRA rank / alpha	16 / 16
Learning rate / schedule	2×10^{-4} / cosine
Quantized optimizer	8-bit AdamW
Training time	316 seconds (5.3 minutes)
Final training loss	0.5093
Adapter size	119.6 MB
GGUF size	2.32 GB

Base and fine-tuned models answered the same 20 HR questions without retrieval. The evaluation assigned one point when an answer contained an HR keyword, used Vietnamese characters, was non-empty, and contained none of eight predefined off-domain/refusal strings; a partial score was possible. Consequently, this is a format and topicality indicator, not semantic fact checking. Greedy generation and per-question wall-clock latency were also recorded.

3.8 Evaluation Protocol

Retrieval and generation were evaluated separately to avoid attributing one component's failure to another. The retrieval set paired ten questions with a gold source-document label. Accuracy@1 and Mean Reciprocal Rank were calculated as

$$\mathrm{Acc@1} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[r_i = 1], \qquad \mathrm{MRR} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{r_i}, \tag{3.2}$$

where r_i is the rank of the labelled document. Dense and hybrid retrieval used the same corpus and queries.

For direct generation, each model received the same 20 questions and no policy context. Decoding was deterministic. The scoring function checked HR-keyword presence, Vietnamese-character frequency, minimum length, and eight literal off-domain/refusal flags. Latency measured one generation call on the H200 evaluation environment. Since answer correctness was not compared with a structured gold answer, manual error analysis is reported alongside the score.

The two evaluations answer different questions. Retrieval metrics measure whether evidence can be found; direct generation metrics measure topical form and speed after adaptation. Neither alone measures end-to-end user trust. A complete future protocol should combine retrieval, source-grounded generation, factual scoring, and human review on a larger held-out set.

4. Results and Discussion

4.1 Retrieval Results

The internal retrieval test contains ten Vietnamese HR questions with manually assigned source documents. Dense search returned the correct top document for six questions, while hybrid retrieval returned it for all ten. MRR improved from 0.72 to 1.00 (Table 4.1). This is evidence for the tested queries, not a population-level accuracy estimate.

Table 4.1: *Retrieval comparison on ten labelled questions.*

Metric	Dense vector	Hybrid RRF
Correct top-1 sources	6/10	**10/10**
Top-1 accuracy	60%	**100%**
Mean Reciprocal Rank	0.72	**1.00**

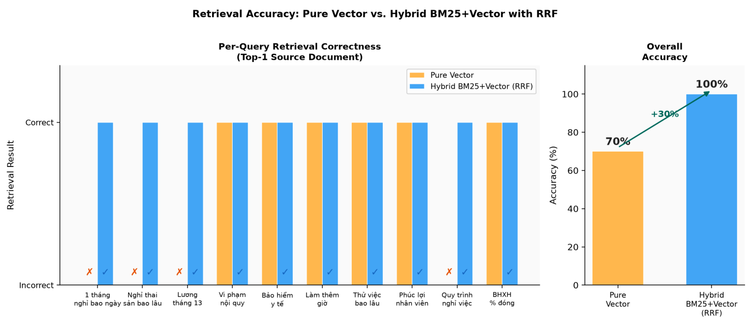


Figure 4.1: *Top-1 retrieval accuracy on the ten-query internal test.*

The largest gains occurred for brief queries such as "one month has how many leave days?" For such wording, BM25 preserved exact leave-related terms while the dense model sometimes selected a salary or general-benefit document. RRF also retained dense evidence for longer paraphrases, so lexical matching did not replace semantic retrieval. The main uncertainty is sample size: ten questions cannot establish robustness across all employee language.

4.2 Fine-Tuning Results

Table 4.2 reports the direct model comparison. Fine-tuning raised the automated score by 0.07 in the report's unrounded calculation, increased the HR-keyword flag from 13 to 14 answers, and reduced mean latency by approximately 77%. Neither model triggered the narrow list of off-domain strings.

Table 4.2: *Direct, retrieval-free evaluation on 20 fixed questions.*

Metric	Base	Fine-tuned	Change
Automated score	0.62	0.70	+0.07
HR-keyword relevant	13/20	14/20	+1
Flag-list matches	0/20	0/20	0
Mean latency	3.9 s	0.9 s	−3.0 s

These numbers show that adaptation changed model behaviour, but qualitative inspection prevents a positive factual-accuracy conclusion. Examples include five days of annual leave, 0.05% social-insurance contribution, 500 VND lunch allowance, and a two-to-three-million-VND answer to a question about recruitment rounds. The flag-list metric recorded none of these as hallucinations because it only searched for words such as "Medicare," "election," and "Japan."

Table 4.3: *Representative fine-tuned outputs requiring factual review.*

Question topic	Observed answer or issue
Annual leave	States a five-day annual maximum without source grounding.
Maternity leave	Answers "60 days per year," confusing maternity and sick leave.
Social insurance	Gives a 0.05% salary contribution.
Recruitment rounds	Answers with a monetary range rather than a number of rounds.
Lunch allowance	Gives 500 VND per working day, an implausible value.
Performance review	Switches to an English list despite a Vietnamese question.

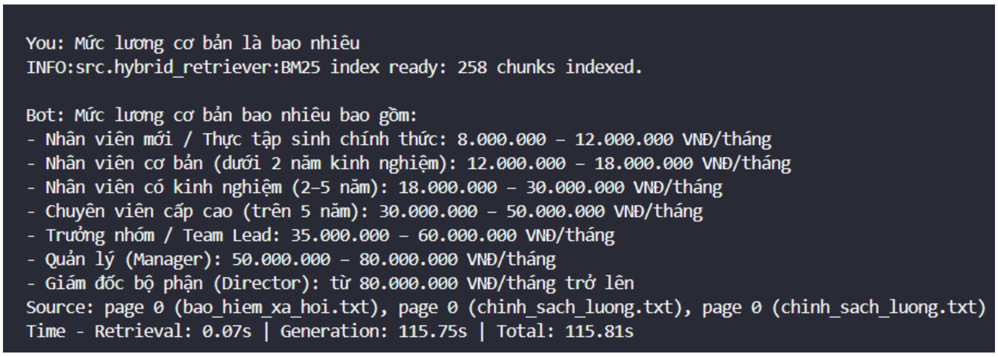
The experiment therefore has a mixed outcome. It is clearly better than the first contaminated-data run: responses are generally on the HR topic and are faster. It is not reliable enough to answer policy questions from parameters alone. Possible causes include synthetic-label errors, contradictory policy paraphrases, limited coverage of numeric facts, and a small model learning surface patterns rather than source-grounded reasoning.

4.3 Role of RAG After Fine-Tuning

The direct evaluation intentionally omitted retrieval, whereas the deployed system supplies policy passages. Fine-tuning and RAG answer different needs: fine-tuning can improve register, response structure, and domain vocabulary; RAG supplies current facts and source attribution.

The observed factual errors support keeping retrieval as the authority. A future comparison should run the base and fine-tuned GGUF models through the identical RAG pipeline and score answers against source-derived references.

Figure 4.2 shows the existing local interface. Retrieval is fast relative to CPU generation, and source labels allow users to verify an answer. This verification path is more defensible than presenting model confidence as policy correctness.



```
You: Mức lương cơ bản là bao nhiêu
INFO:src.hybrid_retriever:BM25 index ready: 258 chunks indexed.

Bot: Mức lương cơ bản bao nhiêu bao gồm:
- Nhân viên mới / Thực tập sinh chính thức: 8.000.000 - 12.000.000 VNĐ/tháng
- Nhân viên cơ bản (dưới 2 năm kinh nghiệm): 12.000.000 - 18.000.000 VNĐ/tháng
- Nhân viên có kinh nghiệm (2-5 năm): 18.000.000 - 30.000.000 VNĐ/tháng
- Chuyên viên cấp cao (trên 5 năm): 30.000.000 - 50.000.000 VNĐ/tháng
- Trưởng nhóm / Team Lead: 35.000.000 - 60.000.000 VNĐ/tháng
- Quản lý (Manager): 50.000.000 - 80.000.000 VNĐ/tháng
- Giám đốc bộ phận (Director): từ 80.000.000 VNĐ/tháng trở lên
Source: page 0 (bao_hiem_xa_hoi.txt), page 0 (chinh_sach_luong.txt), page 0 (chinh_sach_luong.txt)
Time - Retrieval: 0.07s | Generation: 115.75s | Total: 115.81s
```

Figure 4.2: *Local Vietnamese HR chatbot session with retrieved context.*

4.4 Limitations and Threats to Validity

- **Synthetic policies and labels:** the demonstration corpus is not a real company's approved handbook, and generated Q&A pairs were not exhaustively checked against each source sentence.
- **Heuristic validation:** keywords and red-flag strings measure topical form, not entailment, numerical correctness, or legal validity.
- **Small evaluations:** retrieval uses ten labelled queries and direct generation uses twenty questions; confidence intervals and broad generalization claims are not justified.
- **Different execution paths:** the model comparison omits RAG and therefore does not measure the complete deployed chatbot.
- **Vietnamese tokenization:** whitespace BM25 does not segment all compound words correctly.
- **Hardware dependence:** H200 timings demonstrate training efficiency on high-end hardware and do not predict laptop training.

5. Conclusion and Perspective

This project implemented a private Vietnamese HR assistant whose documents, retrieval index, and language model run locally. Hybrid BM25–vector retrieval improved top-1 source selection from 6/10 to 10/10 on a small internal test, showing why lexical and semantic rankings are complementary for short queries.

The H200 experiment generated 4,900 unique Q&A pairs and trained a Phi-3-Mini QLoRA adapter in 5.3 minutes. Fine-tuning improved the automated topicality/ format score and reduced direct inference latency, but manual inspection found incorrect policy facts. The central conclusion is therefore not that fine-tuning solved factual accuracy. Rather, controlled data generation made adaptation more domain-focused, while retrieval remained necessary to provide authoritative, inspectable facts.

Future work should (1) attach each generated answer to an exact supporting span, (2) use source-grounded reference answers and human HR review, (3) compare base and fine-tuned GGUF models inside the same RAG pipeline, (4) adopt Vietnamese word segmentation for BM25, and (5) test with an approved real-company handbook. These steps would turn the current proof of concept into a credible evaluation of practical policy assistance.

A. Reproducibility Map

The full source code, generated data, and raw model outputs are kept in the project repository rather than reproduced as long listings in this thesis. Table A.1 maps each reported result to its local artifact.

Table A.1: *Local evidence artifacts used in this report.*

Artifact	Evidence provided
`data/generated_qa_h200.jsonl`	4,900 accepted questions, answers, source topics, and batch metadata.
`scripts/validate_qa.py`	Reproducible keyword relevance, red-flag, length, and duplicate checks.
`finetune_run.log`	Model ID, hyperparameters, example count, duration, final loss, and export events.
`models/phi3-mini-hr-finetuned/`	LoRA configuration, tokenizer assets, and adapter weights.
`models/phi-3-mini.gguf`	2.32 GB Q4 artifact with HR fine-tune metadata.
`data/eval_scores.csv`	Forty rows containing outputs, flags, scores, and latency.
`data/eval_report.txt`	Base/fine-tuned comparison and answer excerpts.
`src/hybrid_retriever.py`	BM25, vector search, and RRF implementation.

The following commands reproduce the dataset validation and thesis build from the repository root:

```
python scripts/validate_qa.py --input data/generated_qa_h200.jsonl
cd report
compile.bat
pdfinfo thesis.pdf
```

Reproducing inference requires sufficient memory and the Python/Unsloth environment used by the evaluation script. The CSV and text report preserve the completed H200 outputs, so report verification does not require renting a new GPU instance.

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